

CSE422 Lab Project Report

Title: Obesity Level Prediction Using Machine Learning

Group No: 13

Section: 23

Submitted By: Parisa Rahamn

ID: 23101400

Khaled Saifullah Karim

ID: 24341262

Submission Date: 02.01.2026

Table of Contents

Introduction	3
Dataset Description	3
Data Pre-processing	5
Dataset Splitting	6
Model Training and Testing	6
Model Comparison Analysis	7
Conclusion	11

Introduction

Obesity is a major health issue which increases the risk of long term chronic diseases such as hypertension, heart diseases, diabetes. Early identification of obesity level can help to prevent it and raise awareness. The main goal of our project is to predict obesity level of a person based on their physical attributes and lifestyle such as age, height, weight, gender, family history, smoking habits, transportation methods etc. To classify the level of obesity, this project uses supervised machine learning techniques such as KNN, Logistic Regression, Naive Bayes, and Neural Network model. Additionally to determine the hidden pattern of the data, it uses unsupervised learning, K-Means clustering.

Dataset Description

The dataset contains 2111 records, each row representing an individual's physical, nutritional, and lifestyle details. The target feature represents the obesity level of each individual.

Features:

Total Features: 17

Input Features: 16

Target Feature: 1

Problem Type: It's a classification problem. The target feature NObesidad consists of Insufficient_weight, normal_weight, overweight_level_I, overweight_level_II, obesity_type_I, obesity_type_II, obesity_type_III. So it's a multiclass classification problem.

Total Data Points: The given dataset consists of 2111 data points.

Feature Type: There are both quantitative features and categorical features.

Categorical Features: Gender, CALC, FAVC, SCC, SMOKE, family_history_with_overweight, CAEC, MTRANS.

Quantitative Features: Age, Height, Weight, FCVC, NCP, CH2O, FAF, TUE.

Encoding Categorical Variables: As machine learning algorithms cannot process categorical data directly, we used one-hot encoding, label encoding and manual binary mapping to encode categorical variables based on their type.

Label Encoding: Gender, family_history_with_overweight, FAVC, SMOKE, NObeyesdad

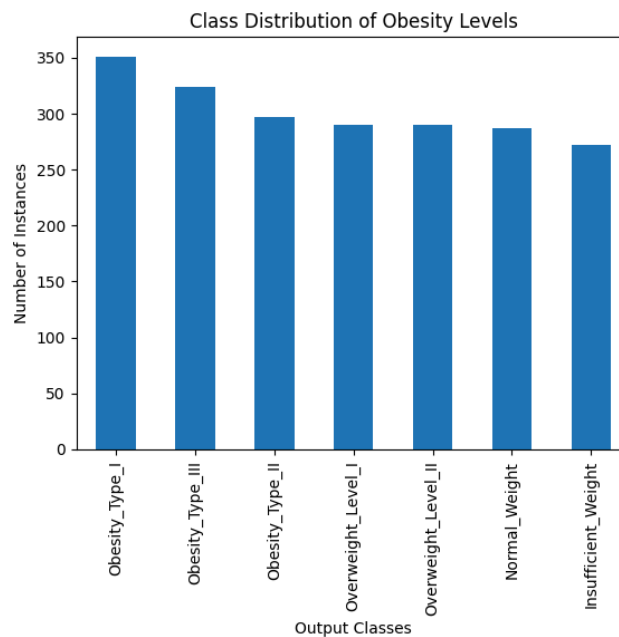
One-Hot Encoding: CAEC, CALC, MTRANS

Manual Binary Mapping: SSC Feature because it contains mixed data type and pandas failed to detect it as an object.

Correlation Analysis: To analyze the relationship between input and output variables, we generated a correlation heatmap using seaborn library.

From the correlation analysis we observed that weight shows a strong positive correlation with obesity level. Height has a moderate relationship with obesity level when combined with weight. Lifestyle related features such as Physical Activity (FAF), Water consumption (CH2O) and Vegetable intake (FCVC) show negative correlations with obesity level, indicating a healthier lifestyle reduces the risk of obesity. The level of obesity cannot be determined fully by a single feature, indicating obesity is influenced by multiple interrelated factors and justify the need of machine learning models to represent the complex relationships.

Imbalance Dataset: The dataset contains seven obesity related classes : Insufficient_weight, normal_weight, overweight_level_I, overweight_level_II, obesity_type_I, obesity_type_II, obesity_type_III. The class distribution of the output feature NObeyesdad is shown using a bar chart.



From the chart, it is observed that, with 351 samples, obesity_type_I has highest number of cases followed by obesity_type_III (324 samples) and obesity_type_II (297 samples). Normal_weight, overweight_level_I, overweight_level_II contain similar number of samples indicating minimal

imbalance among these classes. The Insufficient_weight class has the lowest number of samples, with 272 cases.

Although the classes are not fully balanced but the imbalance is moderate. However, this imbalance can still influence the model prediction toward majority classes. Given this, instead of depending only on model accuracy we focused on evaluation metrics such as precision, recall, confusion matrix, and ROC-AUC.

Data Pre-processing

Before training the machine learning models, the dataset was carefully examined and pre-processed to handle missing values, categorical variables, and feature scaling issues. Proper pre-processing was required to ensure data quality and model reliability.

Fault: Null/Missing Values

Several features in the dataset had missing values, according to an initial analysis. Specifically:

The Gender Feature has 11 missing values

The CH2O Feature has 21 missing values

The family_history_with_overweight has 21 missing values

Solution: Instead of deleting rows, imputation was applied as deleting rows would remove valid information from other features and missing values are not large enough to justify row deletion.

Imputation Strategy Used:

Feature	Type	Imputation Method	Reason
Gender	Categorical	Mode	Preserves most frequent category
family_history_with_overweight	Categorical	Mode	Binary categorical feature
CH2O	Numerical	Mean	Continuous numerical variable

Fault: Categorical Values

The dataset includes several categorical features such as Gender, CALC, FAVC, SCC, SMOKE, family_history_with_overweight, CAEC, and MTRANS. Machine learning algorithms cannot directly process categorical data, making encoding necessary.

Solution:

Encoding Techniques Used:

Feature Type	Encoding	Reason
Binary categorical	Label Encoding	Natural 0/1 mapping
Nominal categorical	One-Hot Encoding	No inherent order
Target variable	Label Encoding	Required for classification

Fault: Feature Scaling

Numerical features such as Age, Height, Weight, CH2O, FAF, and TUE are measured on different scales. This can negatively affect the performance of machine learning models such as KNN and Neural Networks and larger-scale features may dominate model learning.

Solution: To address this issue, StandardScaler from the scikit-learn library was used to standardize the numerical features. Standardization transforms the data so that each feature has:

- A mean of 0
- A standard deviation of 1

Dataset Splitting

The dataset was divided into training, validation and testing sets using stratified sampling. Stratification was applied during all splits to ensure balanced representation of obesity classes in the training, validation, and testing sets.

- Training Set: 70% of total dataset
- Validation Set: 10% of total dataset
- Testing Set: 20% of total dataset

Model Training and Testing

The following supervised learning models were implemented:

1. K-Nearest Neighbors
2. Logistic Regression

- 3. Naive Bayes
- 4. Neural Network

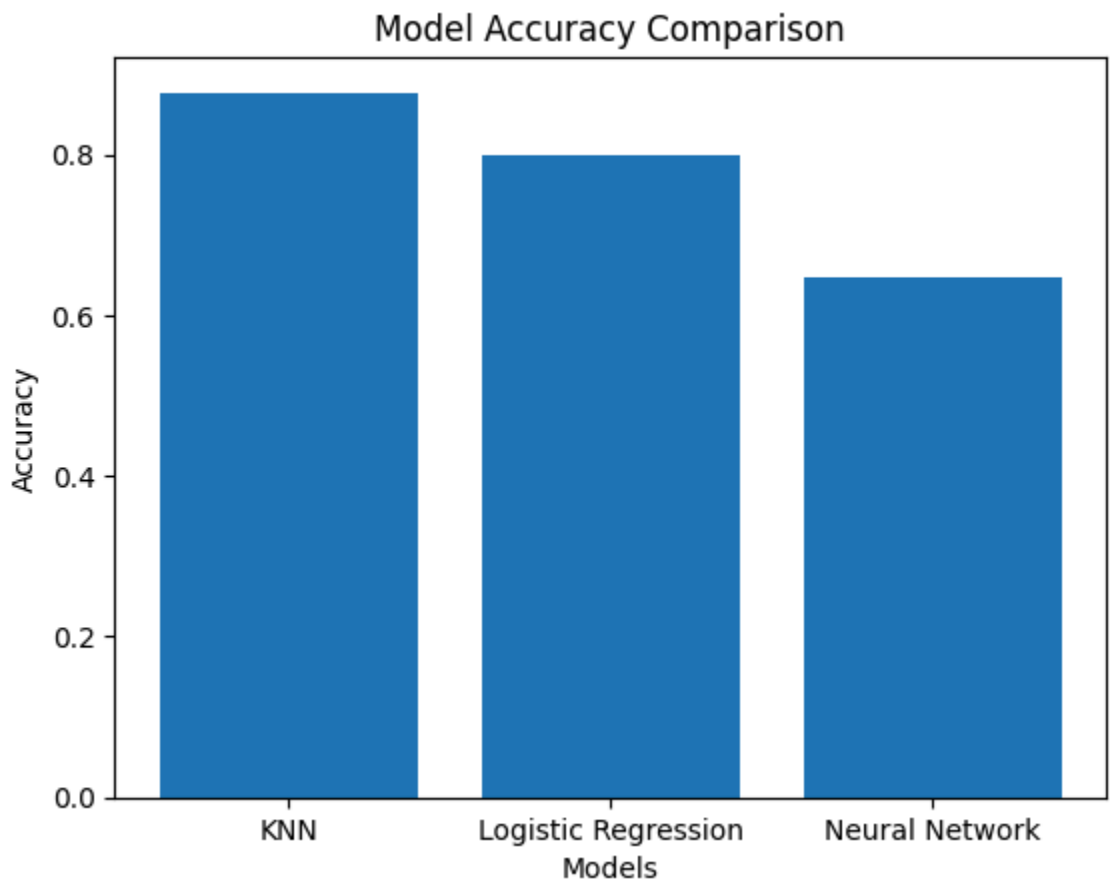
Each model was trained using the training dataset and evaluated using the test dataset.

Unsupervised Learning: To analyze hidden patterns, the dataset was treated as an unsupervised learning problem using K-Means clustering. KMeans effectively grouped individuals in the dataset into 7 clusters with distinct obesity patterns, which could be used for further analysis or solutions. The PCA plot confirms the clusters are reasonably separated in feature space.

Model Comparison Analysis

For classification performance evaluation, the following metrics were used:

Accuracy (visualized using a bar chart)



Precision and Recall

KNN Classification Report

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

0	0.84	0.96	0.90	54
1	0.86	0.62	0.72	58
2	0.94	0.90	0.92	70
3	0.92	0.97	0.94	60
4	0.97	0.97	0.97	65
5	0.80	0.84	0.82	58
6	0.79	0.86	0.83	58
accuracy			0.88	423
macro avg	0.87	0.88	0.87	423
weighted avg	0.88	0.88	0.87	423

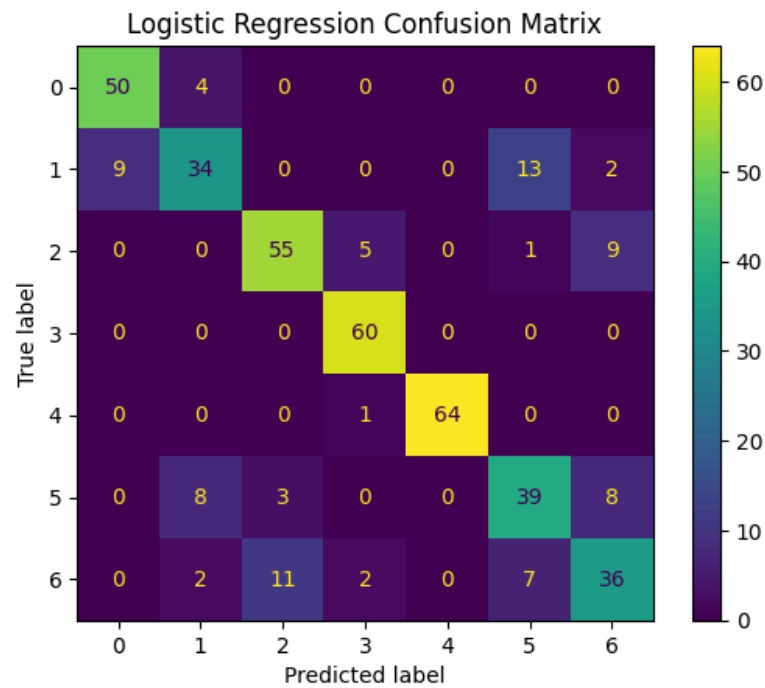
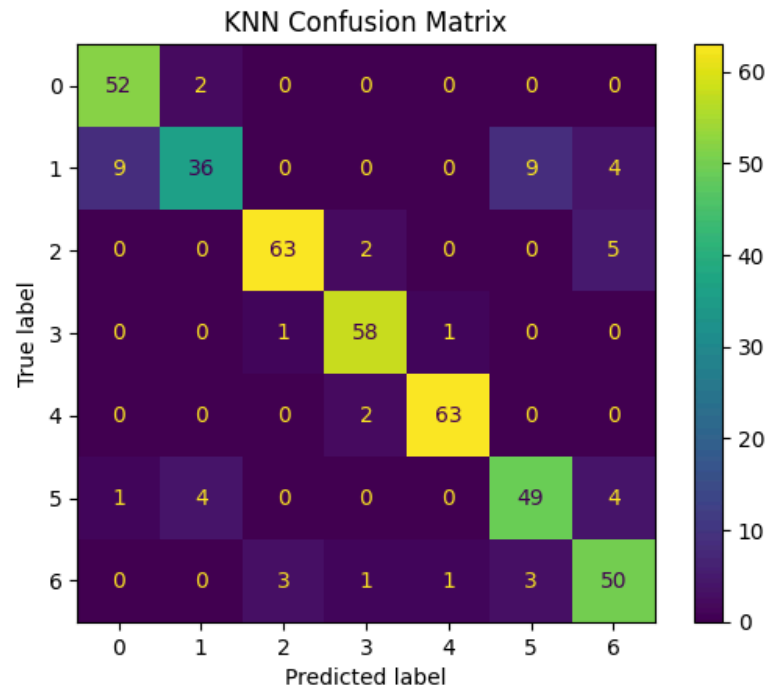
Logistic Regression Classification Report

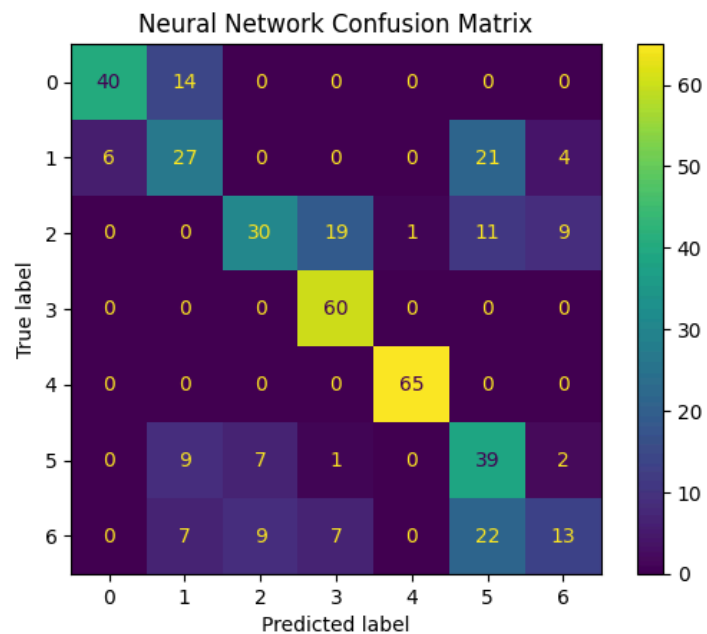
	precision	recall	f1-score	support
0	0.85	0.93	0.88	54
1	0.71	0.59	0.64	58
2	0.80	0.79	0.79	70
3	0.88	1.00	0.94	60
4	1.00	0.98	0.99	65
5	0.65	0.67	0.66	58
6	0.65	0.62	0.64	58
accuracy			0.80	423
macro avg	0.79	0.80	0.79	423
weighted avg	0.79	0.80	0.80	423

Neural Network Classification Report

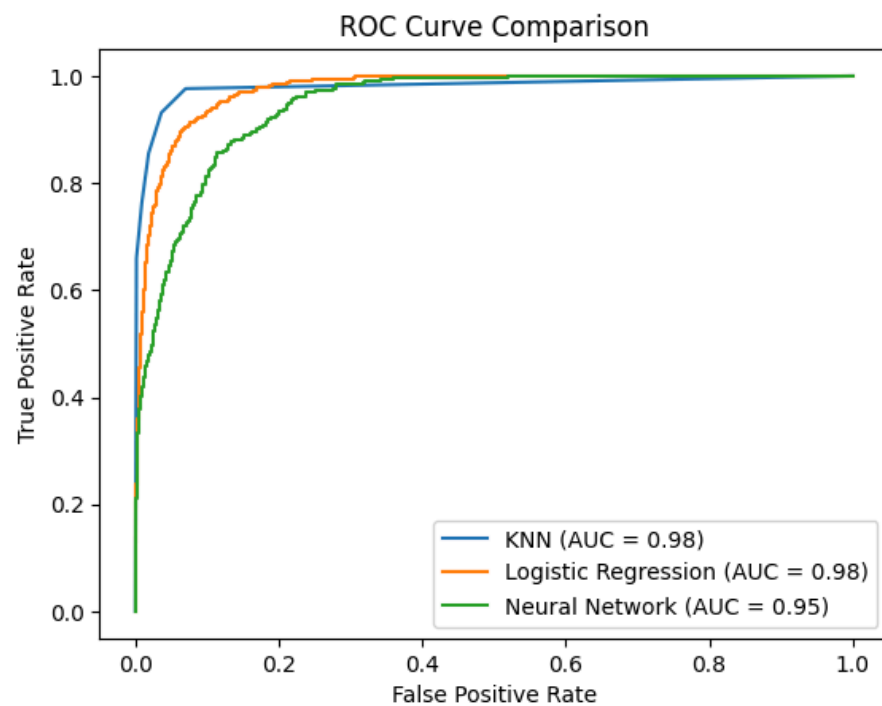
	precision	recall	f1-score	support
0	0.87	0.74	0.80	54
1	0.47	0.47	0.47	58
2	0.65	0.43	0.52	70
3	0.69	1.00	0.82	60
4	0.98	1.00	0.99	65
5	0.42	0.67	0.52	58
6	0.46	0.22	0.30	58
accuracy			0.65	423
macro avg	0.65	0.65	0.63	423
weighted avg	0.65	0.65	0.63	423

Confusion Matrix





ROC Curve and AUC Score



The KNN model achieved the highest accuracy, followed by Logistic Regression and Neural Network models. Naive Bayes performed reasonably well despite its simplicity.

Conclusion

From the experimental results, it can be understood that different machine learning models perform differently depending on the nature of the dataset and feature distribution. In this case, the K-Nearest Neighbors (KNN) classifier outperformed other models, achieving the highest accuracy and demonstrating strong performance across most obesity classes.

This indicates that the obesity dataset contains strong local patterns that are effectively captured by distance-based learning methods.

The features of the dataset can be used to explain the observed outcomes. Since obesity levels are influenced by a combination of lifestyle and physical attributes, similar data points tend to cluster together, which favors KNN. Neural networks, on the other hand, perform quite poorly in this situation since they need larger datasets and meticulous hyperparameter tuning.

Several challenges were encountered during this project, including handling categorical variables, resolving missing values, and ensuring proper feature scaling for distance-based algorithms. It was also difficult to handle class-wise performance imbalance and maintain consistent encoding across training and testing phases. Despite these challenges, suitable model selection and preprocessing came up with acceptable and trustworthy results.